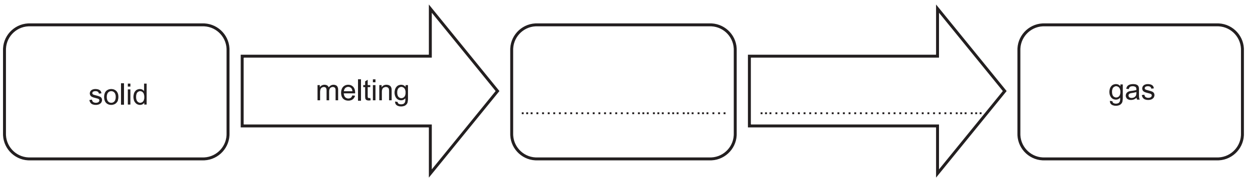
5. The diagram gives some information about what happens when a solid is heated.



(a) **Complete the diagram.** [2]

(b) Tick (✓) the **two** correct statements about a solid. [2]

A solid has the lowest density of the three states of matter.	
The atoms in a solid are in fixed positions.	
The atoms in a solid transfer heat by convection.	
A solid is always a good conductor.	
A solid has atoms that vibrate more as they gain energy.	

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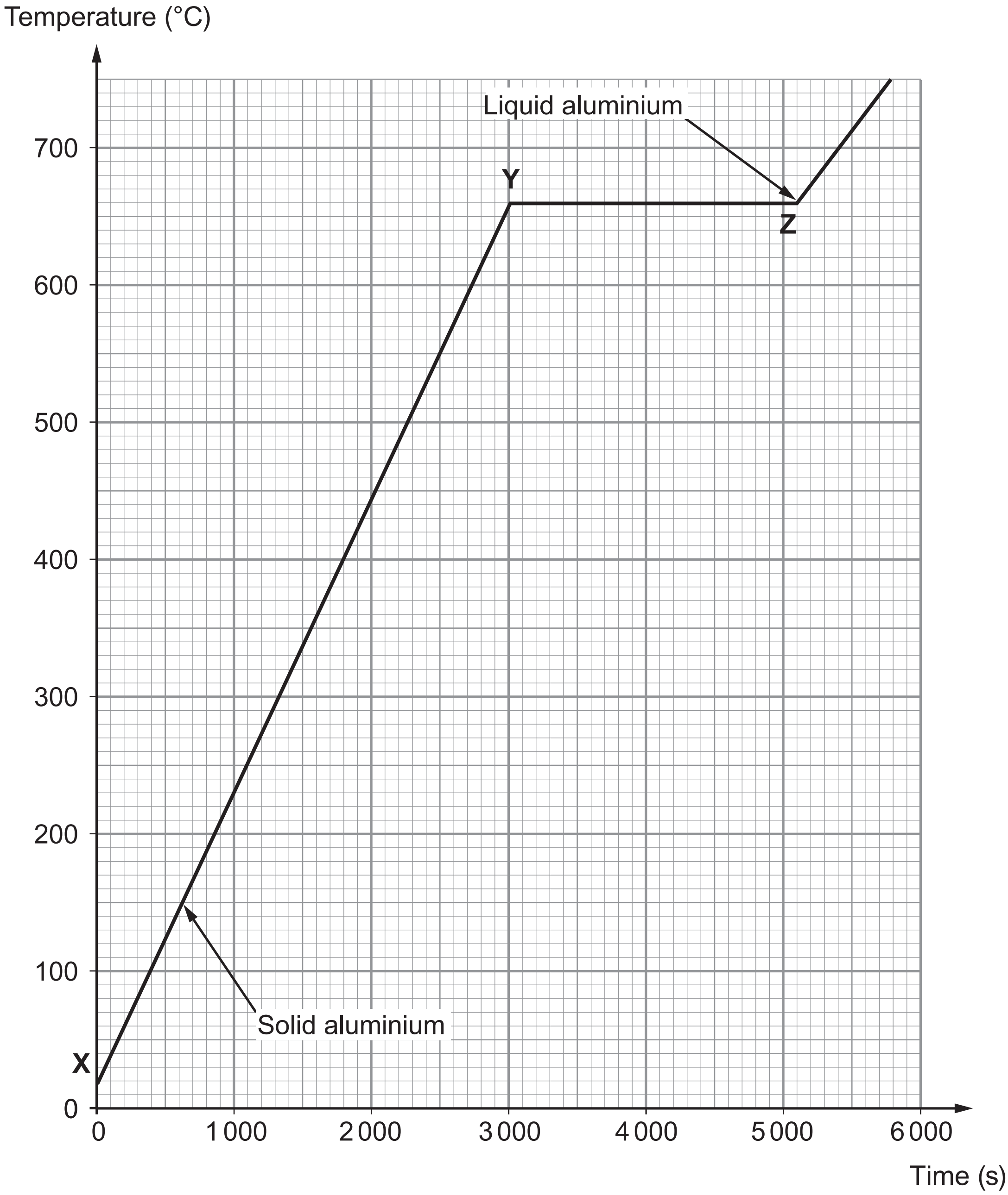
(c) Aluminium cans are frequently recycled. The aluminium cans are collected by local councils as part of household waste. They are sent to a furnace where the cans are heated to melt them. The aluminium is then cooled so that it can be reused for the manufacture of other items. The graph shows how the temperature of the aluminium cans in the furnace changes with time.



(i) State the temperature at which the aluminium cans change from a solid into a liquid. [1]

Temperature = ... °C

(ii) State the time it takes to heat the aluminium cans to their melting point. [1]

Time = ... s

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(d) The heat transferred during the heating process, between points **X** and **Y** is 288 000 000 J.

(i) Use your answer from *(c)*(ii) and the equation:

$$\text{power} = \frac{\text{heat transferred}}{\text{time}}$$

to calculate the power **in kW** of the heater in the furnace. [3]

Power = ... kW

(ii) The temperature change of the aluminium cans during the heating process **X** to **Y** is 640 °C.

Use information given above and the equation:

$$\text{mass} = \frac{\text{heat transfer}}{(\text{specific heat capacity} \times \text{temperature change})}$$

to calculate the mass of aluminium cans that were heated in the furnace.
(Specific heat capacity of aluminium = 900 J/kg °C) [2]

Mass = ... kg

(e) Use an equation from page 2 to calculate the heat transfer required to melt 1 500 kg of aluminium cans from solid to liquid at its melting point. [2]
(Specific latent heat of fusion of aluminium, L = 400 000 J/kg)

Heat transfer = ... J

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